SLANT Contamination Control Plan

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This plan states how SLANT (Solar Limb Absorption of Neutrals in the Thermosphere) is kept clean from parts fabrication through its first weeks in orbit. Each control is identified with its standard. The procedures that carry out the controls are written in Phase 2.

1. Contamination Risks

Three parts of the optical path are sensitive to contamination.

- **Filters.** The five slits are covered by filters that include free-standing zirconium, aluminum and indium films of order 100 nm thick.
- **Slit plate.** Each slit is 25 µm wide.
- **Detector.** The back-illuminated sensor has no window, because any window would absorb the extreme ultraviolet (EUV).

Each sensitive part is exposed to three kinds of contaminant.

- **Molecular.** Polymers outgas hydrocarbons and silicones, which condense on the coldest surfaces they reach. A deposit a few nanometres thick absorbs measurably in the EUV. Ultraviolet light in orbit converts silicone deposits to silica, which heating does not remove.
- **Particulate.** A particle on a slit blocks part of its width, which changes that channel's throughput and angular response. A particle that punctures a film leaves a pinhole, and the pinhole admits unfiltered light.
- **Moisture.** Indium in air forms an oxide 8 to 10 nm thick within days, which then thickens slowly over months at a rate set by the storage conditions [8]. The oxide absorbs in the indium passband.

2. Requirements

Each occultation records the solar signal above the atmosphere and divides every later frame in the pass by it, through the same slit and filter. A throughput loss that is constant over one pass therefore cancels in the measured transmission. Contamination still affects the measurement in two ways.

- **Signal.** A throughput fraction T raises the photon shot noise by a factor 1/√T. Indium aging in orbit is already carried as an accepted risk of about 30 % loss in ten months, a factor of 1.20 in noise.
- **Leak fraction.** A molecular deposit absorbs the in-band EUV more strongly than the longer wavelengths that leak through a filter. The fraction of leaked light in each channel therefore rises as the deposit grows. The retrieval tolerates an out-of-band rejection of at most 3.9×10^{-3}.

The plan allocates to contamination an end-of-life throughput loss for each channel and a limit on the rise in each channel's leak fraction. **Both are allocations to be derived** from the photon

budget's margin and the 3.9×10^{-3} ceiling. The particle inspection criterion for the slits and films is set with them.

3. Materials

Every non-metallic material inside the housing, or with a view of the aperture, meets the ASTM E595 screening levels used by NASA: total mass loss (TML) of at most 1.0 % and collected volatile condensable material (CVCM) of at most 0.10 %. Both are measured after 24 hours at 125 °C in vacuum [1, 2].

A materials list records each material, its use, its mass and its E595 values. The list includes:

- the black anodize on the baffle cone;
- the cure-in-place gap filler under the processor;
- harness insulation and connector inserts;
- staking and bonding adhesives, including the epoxy Luxel uses to bond the films.

No silicone is used where it has a view of the filters or the detector. The TI space-enhanced parts on the carrier board are qualified to E595 by TI. The commercial parts, including the Raspberry Pi Compute Module 5 (CM5), are not, and the unit-level bake-out (Section 5) reduces their outgassing.

4. Fabrication and Cleaning

- **Housing parts.** Precision-cleaned after machining and finishing, then bagged in low-outgassing film.
- **Slit plates.** Cleaned by the etching vendor to the level Luxel requires for its vacuum bond and bake. Luxel's incoming requirement governs.
- **Carrier board and harness.** Built to J-STD-001, with flux removed and cleanliness verified as that standard requires.
- **Parts-level bake.** Finished housing parts and the harness are baked in vacuum at 100 °C for 100 hours. This is the practice of NASA's Space Environments Complex for painted metals and cable sets [5].

5. Bake-Out

The assembled unit is baked in vacuum with the filter cartridge removed, so the films are not present while outgassing rates are highest. The bake follows MSFC-SPEC-1238 [3]:

- The hardware is held at the highest temperature that does not endanger it, and at least 10 °C above its maximum on-orbit operating temperature.
- Chamber pressure stays below 1×10^{-5} Torr.
- A temperature-controlled quartz crystal microbalance (TQCM) is held 10 °C below the minimum orbital temperature of the coldest optical element.

- The bake is complete when the TQCM deposition rate, averaged over 36 hours, is 1 Hz/h or less.

ASTM E2900 gives equivalent bake methods [4].

Temperature. The camera head limits the bake temperature. XIMEA rates its xiJ cameras for storage from −25 to +60 °C [6], the lowest maximum among the parts whose ratings are on file (the CM5 is rated to operate up to +85 °C [7]). The unit is therefore baked at +55 °C, which leaves 5 °C for chamber control. The baseplate's survival range tops out at +50 °C, and its limit while acquiring is +20 °C. The +55 °C bake meets the 10 °C margin of MSFC-SPEC-1238 if no internal part runs above +45 °C while acquiring. The detector-heating measurement on the engineering unit checks this. If XIMEA confirms a higher limit for the flight head, the bake temperature rises to that limit, less 5 °C.

The bake runs at UW-Madison in months 7-8. Carthage's vacuum chamber is used for the engineering unit.

6. Integration and Handling

- **Assembly.** Units are assembled at Carthage on a laminar-flow bench. Its class has not been measured. Before flight assembly, particle counts are taken at the work surface and the bench is classified to ISO 14644-1.
- **Clean room.** The filter cartridge is installed after the bake in the clean room at the UW-Madison Space Science and Engineering Center (SSEC). Any later opening of the unit is also done there.
- **Garments and ESD.** Powder-free gloves and clean-room garments are worn throughout, with electrostatic discharge control.
- **Filters.** Kept in their shipping containers under dry nitrogen until installation, and handled to Luxel's instructions. Luxel's storage instructions for indium are to be requested.
- **Ground-only aperture cover.** A remove-before-flight cover with a filtered vent is fitted whenever the unit is outside a clean room. It keeps particles and direct airflow off the films, and it is removed before the unit is enclosed in the fairing.
- **Storage and shipping.** The unit is double-bagged in low-outgassing antistatic film, purged with dry nitrogen, and packed with a humidity indicator.

7. Launch

SLANT launches vented, in air, with no flight cover.

- **Vent paths.** Grooves with a sideways step let the cavity behind the films depressurize, and the step leaves no straight line of sight into the cavity. Wall vents carry outgassing from the electronics volume out of the housing, and a skirt around the filter cartridge keeps that outgassing from crossing the films.
- **Pressure across the films.** A hand calculation for the vents as drawn gives a peak pressure difference across the films of 0.02 Torr at sea level and 0.09 to 0.21 Torr near 10 km altitude.

These figures assume fairing depressurization rates up to 4.5 kPa/s. Luxel's limit is 1 Torr.

- **Indium survival.** Luxel advises that the indium film would likely need a vacuum launch. Because the pressure difference is within Luxel's limit, the open question is whether the film survives launch acoustics and vibration in air. An acoustic and random-vibration test of a flight-like indium film assembly early in Phase 2 answers it. If that film fails, a sealed one-shot cover is the fallback.
- **Fairing cleanliness.** From cover removal until launch, the aperture is open to the host and fairing environment. The launch provider specifies that environment, and SLANT's cleanliness request goes into the host interface control document.

8. Early Orbit

Outgassing from the unit and the host is highest in the first weeks, and the vapor condenses on the coldest surface it reaches. SLANT has no detector cooler. When the camera is powered, its own dissipation keeps the detector at or above the baseplate temperature. When the unit is off between passes, the detector cools toward the baseplate.

For the first weeks, the host keeps SLANT powered between passes as far as its power budget allows, which holds the detector above the surrounding housing. The length of this period is set from the outgassing decay recorded during the bake. The thermal analysis will identify the coldest surface with a view of the film cavity. If that surface is a filter film, this section is revised.

9. Verification and Monitoring

- The materials list is checked against E595 data before parts are ordered.
- The slits and films are inspected and photographed under magnification before the cartridge is closed.
- The TQCM record shows that the bake is complete.
- Witness plates for molecular and particulate fallout accompany the unit from the bake to delivery, and are measured at delivery.
- In orbit, the solar signal recorded above the atmosphere in each pass gives each channel's throughput trend. The signal recorded below 105 km in each occultation gives each channel's leak fraction. Both come from data SLANT already records.

10. Responsibilities

- **Kevin Crosby:** owns the plan and approves the materials list.
- **Fred Best:** bake-out and thermal vacuum at UW-Madison.
- **Maya Griffin:** carries out the bake-out and thermal vacuum with Best, and puts the material callouts on the drawings.
- **Owen Bonnett:** assembly, filter handling and the inspection records.
- **Semaje Farmer:** workmanship to J-STD-001 and cleanliness of the soldered assemblies.

References

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